Convergence Tests

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: The ratio and comparison tests**

Use the ratio test to determine whether $\sum_{n=1}^{\infty} \frac{n}{2^n}$ converges.

$\lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty} \frac{n+1}{2n} = \frac{1}{2} < 1$: converges.

2 **[Foundational]** • **Target: The ratio and comparison tests**

Use the ratio test on $\sum_{n=1}^{\infty} \frac{n!}{n^n}$.

$\frac{a_{n+1}}{a_n} = \frac{n^n}{(n+1)^n} = \left(\frac{n}{n+1}\right)^n \to \frac{1}{e} < 1$: converges.

3 **[Foundational]** • **Target: The ratio and comparison tests**

Use the comparison test to determine whether $\sum_{n=1}^{\infty} \frac{1}{n^2+3}$ converges, by comparing to $\sum \frac{1}{n^2}$.

$0 < \frac{1}{n^2+3} < \frac{1}{n^2}$ for all $n \geq 1$, and $\sum \frac{1}{n^2}$ is a convergent p-series ($p = 2 > 1$), so by comparison the given series converges.

4 **[Foundational]** • **Target: Alternating series and the p-series test**

Determine whether each p-series converges or diverges:

a $\sum_{n=1}^{\infty} \frac{1}{n^3}$ Converges ($p = 3 > 1$).

b $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}}$ Diverges ($p = \frac{1}{2} \leq 1$).

(a) Converges ($p = 3 > 1$).; (b) Diverges ($p = \frac{1}{2} \leq 1$).

5 **[Foundational]** • **Target: Alternating series and the p-series test**

Determine whether $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n}$ converges, and classify it as absolutely convergent, conditionally convergent, or divergent.

The terms $\frac{1}{n}$ decrease to 0, so by the alternating series test the series converges. But $\sum \frac{1}{n}$ (absolute values) diverges — so the series is conditionally convergent.

6 **[Foundational]** • **Target: The integral test**

Use the integral test to determine whether $\sum_{n=1}^{\infty} \frac{1}{n \ln n}$ (starting at $n = 2$) converges or diverges, using $\int_2^{\infty} \frac{1}{x \ln x}\,dx$.

Let $u = \ln x$: $\int \frac{1}{x \ln x} dx = \ln|\ln x| + C$. $\int_2^{\infty} \frac{dx}{x \ln x} = \lim_{b\to\infty} [\ln(\ln x)]_2^b = \infty$: the integral diverges, so by the integral test the series diverges.

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7 [Foundational] • Target: The integral test

Use the integral test to confirm that $\sum_{n=1}^{\infty} \frac{1}{n^2}$ converges, using $\int_1^{\infty} \frac{1}{x^2}\,dx$.

$\int_1^{\infty} \frac{1}{x^2}dx = \lim_{b\to\infty} \left[-\frac{1}{x}\right]_1^b = 1$, a finite value, so by the integral test the series converges.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Choosing the right test

For each series, name the most efficient test to apply (ratio, comparison, p-series, or alternating series), without fully evaluating:

a $\sum \frac{2^n}{n!}$ Ratio test (factorial in denominator is a strong signal).

b $\sum \frac{(-1)^n}{\sqrt{n}}$ Alternating series test (terms decrease to 0).

c $\sum \frac{1}{n^4}$ p-series test directly ($p = 4 > 1$: converges).

(a) Ratio test (factorial in denominator is a strong signal).; (b) Alternating series test (terms decrease to 0).; (c) p-series test directly ($p = 4 > 1$: converges).

9 [Intermediate] • Target: The limit comparison test

Use the limit comparison test to determine whether $\sum_{n=1}^{\infty} \frac{n+1}{n^3+2}$ converges, comparing to $\sum \frac{1}{n^2}$.

$\lim_{n\to\infty} \frac{(n+1)/(n^3+2)}{1/n^2} = \lim_{n\to\infty} \frac{n^3+n^2}{n^3+2} = 1$, a finite positive value. Since $\sum \frac{1}{n^2}$ converges ($p = 2 > 1$), by the limit comparison test the given series also converges.

10 [Intermediate] • Target: The limit comparison test

Use the limit comparison test on $\sum_{n=1}^{\infty} \frac{3n^2-1}{n^4+2n}$, comparing to $\sum \frac{1}{n^2}$.

$\lim_{n\to\infty} \frac{(3n^2-1)/(n^4+2n)}{1/n^2} = \lim_{n\to\infty} \frac{3n^4-n^2}{n^4+2n} = 3$, finite and positive. Since $\sum \frac{1}{n^2}$ converges, so does the given series.

11 [Intermediate] • Target: The root test

Use the root test to determine whether $\sum_{n=1}^{\infty} \left(\frac{n}{2n+1}\right)^n$ converges.

$\lim_{n\to\infty} \sqrt[n]{\left|\left(\frac{n}{2n+1}\right)^n\right|} = \lim_{n\to\infty} \frac{n}{2n+1} = \frac{1}{2} < 1$: converges.

12 [Intermediate] • Target: The root test

Use the root test on $\sum_{n=1}^{\infty} \frac{3^n}{n^n}$.

$\lim_{n\to\infty} \sqrt[n]{\frac{3^n}{n^n}} = \lim_{n\to\infty} \frac{3}{n} = 0 < 1$: converges.

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13 [Intermediate] • Target: Absolute versus conditional convergence

Determine whether $\sum_{n=1}^{\infty} \frac{(-1)^n}{n^2}$ is absolutely convergent, conditionally convergent, or divergent.

Taking absolute values: $\sum \frac{1}{n^2}$ is a convergent p-series ($p = 2 > 1$). Since the series of absolute values converges, the original series is absolutely convergent.

14 [Intermediate] • Target: Estimating the sum of an alternating series

The alternating series $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2}$ converges by the alternating series test. Use the fact that the error after n terms is bounded by the first omitted term to estimate how many terms are needed for the partial sum to be within 0.001 of the true sum.

Error bound: $|S - S_n| \le a_{n+1} = \frac{1}{(n+1)^2}$. Need $\frac{1}{(n+1)^2} < 0.001$: $(n+1)^2 > 1000$, so $n + 1 > 31.6$, meaning $n \ge 32$ terms suffice.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: A combined convergence-test challenge • Cross-Lesson: 7.01 Integration by Substitution

Determine whether $\sum_{n=1}^{\infty} \frac{n^2}{3^n}$ converges, using the ratio test.

$\left|\frac{a_{n+1}}{a_n}\right| = \frac{(n+1)^2}{3^{n+1}} \cdot \frac{3^n}{n^2} = \frac{1}{3}\left(\frac{n+1}{n}\right)^2 \to \frac{1}{3} < 1$: converges.

16 [Challenging] • Target: A combined convergence-test challenge • Cross-Lesson: 7.02 Integration by Parts

Determine whether $\sum_{n=2}^{\infty} \frac{1}{n(\ln n)^2}$ converges, using the integral test with $u = \ln x$.

$\int_2^{\infty} \frac{1}{x(\ln x)^2}\,dx$: with $u = \ln x$, $du = \frac{1}{x}dx$: $\int \frac{1}{u^2}\,du = -\frac{1}{u} + C$. Evaluating: $\lim_{b\to\infty}\left[-\frac{1}{\ln x}\right]_2^b = 0 - \left(-\frac{1}{\ln 2}\right) = \frac{1}{\ln 2}$, a finite value: the integral converges, so by the integral test the series converges.

17 [Challenging] • Target: A combined convergence-test challenge • Cross-Lesson: 8.01 Area Between Curves

A student claims that since $\lim_{n\to\infty} \frac{1}{n} = 0$, the series $\sum_{n=1}^{\infty} \frac{1}{n}$ must converge. Explain what is wrong with this reasoning, and state the correct necessary (but not sufficient) condition that the nth-term test actually provides, using the harmonic series as a counterexample.

The reasoning confuses a necessary condition with a sufficient one. The nth-term test only says: IF $\lim_{n\to\infty} a_n \neq 0$, THEN the series diverges. It says nothing when the limit IS 0 — that case is inconclusive. The harmonic series $\sum \frac{1}{n}$ is the standard counterexample: its terms go to 0, yet the series diverges (proven via the integral test or the comparison/grouping argument), showing that $\lim a_n = 0$ alone never guarantees convergence.

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18 [Challenging] • Target: A multi-step convergence investigation • Cross-Lesson: 8.02 Volumes of Revolution

For the series $\sum_{n=1}^{\infty} \frac{n^3+2n}{2n^5-n+1}$, first find the nth-term limit to check the necessary condition, then use the limit comparison test with $\sum \frac{1}{n^2}$ to determine whether the series converges.

nth-term limit: $\lim_{n\to\infty} \frac{n^3+2n}{2n^5-n+1} = 0$ (inconclusive by itself). Limit comparison with $\frac{1}{n^2}$: $\lim_{n\to\infty} \frac{(n^3+2n)/(2n^5-n+1)}{1/n^2} = \lim_{n\to\infty} \frac{n^5+2n^3}{2n^5-n+1} = \frac{1}{2}$, finite and positive. Since $\sum \frac{1}{n^2}$ converges ($p = 2 > 1$), by the limit comparison test the given series also converges.

19 [Challenging] • Target: A multi-step convergence investigation • Cross-Lesson: 8.03 Arc Length and Improper Integrals

A quality-control engineer models cumulative defect probability using $\sum_{n=1}^{\infty} \frac{1}{n \cdot 2^n}$. Use the ratio test to confirm this series converges, and briefly explain (in one sentence) why a convergent bound like this is useful for guaranteeing that a manufacturing process's total defect risk stays finite over infinitely many production runs.

$\left|\frac{a_{n+1}}{a_n}\right| = \frac{n \cdot 2^n}{(n+1) \cdot 2^{n+1}} = \frac{n}{2(n+1)} \to \frac{1}{2} < 1$: converges. Practically, a convergent bound guarantees the cumulative defect probability approaches a finite ceiling rather than growing without bound as more production runs occur, which is essential for meaningful long-run risk assessment.

20 [Challenging] • Target: A multi-step convergence investigation • Cross-Lesson: 9.01 Separable Differential Equations

Level 1 · Foundations Use the Divergence Test to determine whether $\sum_{n=1}^{\infty} \cos\left(\frac{1}{n}\right)$ converges or diverges.

Compute the limit of the general term as $n \to \infty$: $\lim_{n\to\infty} a_n = \lim_{n\to\infty} \cos\left(\frac{1}{n}\right) = \cos\left(\lim_{n\to\infty} \frac{1}{n}\right) = \cos(0) = 1$ Because $\lim_{n\to\infty} a_n = 1e0$, the series diverges by the Divergence Test.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: The ratio and comparison tests

Series $\sum_{n=1}^{\infty} (-1)^{n-1}/\sqrt{n}$ converges by alternating series test. Is it absolutely convergent?

a. Yes, $p = 1/2 < 1$ diverges

b. **No, $\sum 1/\sqrt{n}$ diverges ($p = 1/2 \le 1$) ✓**

c. Yes, $\sum 1/\sqrt{n}$ converges

d. Cannot determine

$p = 1/2 \le 1 \Rightarrow \sum 1/\sqrt{n}$ diverges: only conditionally convergent.

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22 [Foundational] • Target: Alternating series and the p-series test

Integral test convergence of $\sum_{n=1}^{\infty} 1/n(\ln n)^2$ (from $n=2$) gives sum approximately:

a. **converges ✓**

b. diverges

c. equals e

d. equals π^2

$\int_2^{\infty} dx/(x(\ln x)^2) = 1/\ln 2 < \infty \Rightarrow$ converges.

23 [Foundational] • Target: The integral test

Which test determines $\sum n^2/e^n$? Answer: by ratio test $L =$

a. e

b. 1

c. $1/e < 1$ ✓

d. e^2

$L = \lim (n+1)^2/e^{n+1} \cdot e^n/n^2 = 1/e < 1 \Rightarrow$ converges.

24 [Foundational] • Target: Choosing the right test

Series $\sum_{n=1}^{\infty} n \cdot (1/2)^n$: ratio test gives $L =$

a. 1

b. 2

c. $1/2$ ✓

d. $n/2$

$|a_{n+1}/a_n| = (n+1)/(2n) \to 1/2 < 1 \Rightarrow$ converges.

25 [Foundational] • Target: The limit comparison test

Use the integral test to show $\sum_{n=2}^{\infty} \frac{1}{n(\ln n)^2}$ converges. The integral $\int_2^{\infty} \frac{dx}{x(\ln x)^2} =$

a. ∞

b. π

c. $\ln 2$

d. $1/\ln 2$ ✓

$u = \ln x$: $\int_{\ln 2}^{\infty} u^{-2} du = 1/\ln 2 < \infty$.

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26 [Foundational] • Target: The root test

The ratio test gives $L = 1$: the test is:

a. **inconclusive ✓**

b. divergent

c. convergent

d. absolutely convergent

Applying Convergence Tests and Power Series principles, Ratio test inconclusive at $L = 1$. Therefore, the verified correct option is inconclusive.

27 [Foundational] • Target: Absolute versus conditional convergence

Limit comparison: $a_n = 1/(n^2 + 1)$, compare to $b_n = 1/n^2$. $\lim a_n/b_n =$

a. 0

b. 1/2

c. ∞

d. 1 ✓

Applying Convergence Tests and Power Series principles, $(n^2)/(n^2 + 1) \to 1$. Therefore, the verified correct option is 1.

28 [Foundational] • Target: Estimating the sum of an alternating series

Find the radius of convergence of $\sum_{n=0}^{\infty} \frac{(2x)^n}{n!}$.

a. 0

b. 2

c. ∞ ✓

d. $\frac{1}{2}$

$\left|\frac{a_{n+1}}{a_n}\right| = \frac{2|x|}{n+1} \to 0$ for every x, so $R = \infty$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: A combined convergence-test challenge

Convergence of $\sum n^2/3^n$: by ratio test, $L =$

a. 1

b. 1/3 ✓

c. 3

d. 0

$|(n+1)^2/3^{n+1} \cdot 3^n/n^2| \to 1/3 < 1 \Rightarrow$ converges.

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30 [Intermediate] • Target: A multi-step convergence investigation

Alternating test: $a_n = 1/n!$: does $a_n \to 0$? Does $a_{n+1} < a_n$?

a. Neither

b. **Both $a_n \to 0$ and $a_{n+1} < a_n$ ✓**

c. $a_n \to 0$ only

d. $a_{n+1} > a_n$

$1/n! \to 0$ and is decreasing: both conditions met.

31 [Intermediate] • Target: The ratio and comparison tests

Comparison: if $\sum b_n$ diverges and $a_n \geq b_n > 0$, then $\sum a_n$:

a. converges

b. equals $\sum b_n$

c. is inconclusive

d. **diverges ✓**

Comparison: dominated from below by divergent ■ diverges.

32 [Intermediate] • Target: Alternating series and the p-series test

The integral $\int_0^1 e^{-x^2} dx$ (error function) — the exact value cannot be expressed in elementary functions. Using the series, approximate as $\sum \frac{(-1)^n}{n!\,(2n+1)}$. First two terms:

a. $1 - 1/6 \approx 0.833$

b. $1 - 1/3 \approx 0.667$ ✓

c. $1 + 1/3 \approx 1.333$

d. 0

$\int_0^1 e^{-x^2} dx \approx 1 - 1/3 + \cdots \approx 0.667.$

33 [Intermediate] • Target: The integral test

Comparison: $0 \leq a_n \leq 1/n^2$. Since $\sum 1/n^2$ converges, $\sum a_n$:

a. diverges

b. may diverge

c. **converges ✓**

d. oscillates

Direct comparison with convergent series.

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34 [Intermediate] • Target: Choosing the right test

Bessel function $J_0(x) = \sum_{m=0}^{\infty} \frac{(-1)^m}{(m!)^2}\left(\frac{x}{2}\right)^{2m}$. Radius of convergence:

a. 1

b. ∞ ✓

c. π

d. 2

Ratio test: $L = 0 \Rightarrow R = \infty$.

35 [Intermediate] • Target: The limit comparison test

Absolute convergence implies conditional convergence. The converse:

a. is also true

b. **is false (counterexample: alternating harmonic series)** ✓

c. requires $a_n \geq 0$

d. is true for power series

Conditional does not imply absolute.

36 [Intermediate] • Target: The root test

Consider the power series $\sum_{n=0}^{\infty} c_n(x-a)^n$. If the radius of convergence is $R = 3$ and the center of the series is $a = 2$, which of the following intervals represents the values of x for which the series converges?

a. $(-1, 5)$ ✓

b. $[-1, 5]$

c. $(-1, 5]$

d. $[-1, 5)$

The radius of convergence R of a power series $\sum_{n=0}^{\infty} c_n(x-a)^n$ indicates that the series converges absolutely for $|x-a|R$. The center of the series is a. In this problem, we are given $R = 3$ and $a = 2$. Thus, the series converges for $|x-2| < 3$. This inequality can be rewritten as $-3 < x-2 < 3$. Adding 2 to all parts of the inequality, we get $-3+2 < x < 3+2$, which simplifies to $-1 < x < 5$. This interval is written in open interval notation as $(-1, 5)$. The convergence at the endpoints.

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37 [Intermediate] • Target: Absolute versus conditional convergence

Apply the ratio test to $\sum_{n=1}^{\infty} \frac{6^n}{n!}$. What is $L = \lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right|$?

a. **0** ✓

b. ∞

c. 1

d. 6

$\left|\frac{a_{n+1}}{a_n}\right| = \frac{6}{n+1} \to 0$ as $n \to \infty$. Since $L = 0 < 1$, the series converges (for every x).

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Estimating the sum of an alternating series • Cross-Lesson: 8.02 Volumes of Revolution

Consider the power series $\sum_{n=1}^{\infty} \frac{(2x-1)^n}{n \cdot 3^n}$. For what values of x does this series converge conditionally?

a. $-1 \le x < 2$

b. $-1 < x \le 2$ ✓

c. $-1 \le x \le 2$

d. $-1 < x < 2$

Let the given power series be $\sum_{n=1}^{\infty} a_n$, where $a_n = \frac{(2x-1)^n}{n \cdot 3^n}$. We will use the Ratio Test to find the interval of convergence. $\lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty} \left|\frac{(2x-1)^{n+1}}{(n+1)\cdot 3^{n+1}} \cdot \frac{n \cdot 3^n}{(2x-1)^n}\right|$.

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39 [Challenging] • Target: A combined convergence-test challenge • Cross-Lesson: 8.03 Arc Length and Improper Integrals

Consider the power series $\sum_{n=1}^{\infty} \frac{(x-2)^n}{n^2 3^n}$. This series defines a function $f(x)$ over its interval of convergence. Let I be the interval of convergence for this power series. Determine which of the following statements is TRUE about $f(x)$ and its behavior at the endpoints of I.

a. The radius of convergence is 3, and the series converges at both endpoints of the interval of convergence. ✓

b. The radius of convergence is 3, and the series diverges at the left endpoint and converges at the right endpoint.

c. The radius of convergence is $1/3$, and the series converges at both endpoints of the interval of convergence.

d. The radius of convergence is $1/3$, and the series diverges at both endpoints of the interval of convergence.

We use the Ratio Test to find the radius of convergence. Let $a_n = \frac{(x-2)^n}{n^2 3^n}$. Then, $L = \lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty} \left|\frac{(x-2)^{n+1}}{(n+1)^2 3^{n+1}} \cdot \frac{n^2 3^n}{(x-2)^n}\right| \; L = \lim_{n\to\infty} \left|\frac{x-2}{3} \cdot \frac{n^2}{(n+1)^2}\right| = \left|\frac{x-2}{3}\right| \lim_{n\to\infty} \left(\frac{n}{n+1}\right)^2 = \left|\frac{x-2}{3}\right| \cdot 1$ For convergence, we need $L1$.

40 [Challenging] • Target: A multi-step convergence investigation • Cross-Lesson: 9.01 Separable Differential Equations

Find the interval of convergence of $\sum \frac{(x-1)^n}{n3^n}$:

a. $[-2, 4)$ ✓

b. $(-2, 4)$

c. $(-2, 4]$

d. $[-2, 4]$

By the Ratio Test, $R = 3$ centered at $x = 1$. At $x = -2$ we get the convergent alternating harmonic series $\sum \frac{(-1)^n}{n}$, and at $x = 4$ the divergent harmonic series $\sum \frac{1}{n}$, so the interval is $[-2, 4)$.

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41 [Challenging] • Target: The ratio and comparison tests • Cross-Lesson: 1.01 Precalculus Review for Calculus

The error function $\operatorname{erf}(x) = \frac{2}{\sqrt{\pi}}\int_0^x e^{-t^2}dt$. Using $e^{-t^2} = \sum(-1)^n t^{2n}/n!$:

a. $\operatorname{erf}(x) = \sum x^n/n!$

b. $\operatorname{erf}(x) = \frac{2}{\sqrt{\pi}}\sum_{n=0}^{\infty}\frac{(-1)^n x^{2n+1}}{n!\,(2n+1)}$ ✓

c. $\operatorname{erf}(x) = e^{-x^2}$

d. $\operatorname{erf}(x) = 2x/\sqrt{\pi}$

Applying Convergence Tests and Power Series principles, Integrate term-by-term. Therefore, the verified correct option is $\operatorname{erf}(x) = \frac{2}{\sqrt{\pi}}\sum_{n=0}^{\infty}\frac{(-1)^n x^{2n+1}}{n!\,(2n+1)}$.

42 [Challenging] • Target: Alternating series and the p-series test • Cross-Lesson: 10.01 Calculus with Parametric and Polar Curves

For a power series $\sum c_n x^n$, the radius of convergence R satisfies $1/R =$

a. $\limsup|c_n|^{1/n}$ ✓

b. $\limsup|c_{n+1}/c_n|$

c. $\sup|c_n|$

d. $|c_0|$

Applying Convergence Tests and Power Series principles, Cauchy-Hadamard formula. Therefore, the verified correct option is $\limsup|c_n|^{1/n}$.

43 [Challenging] • Target: The integral test • Cross-Lesson: 11.01 Sequences and Series

Integral test: $\sum_{n=1}^{\infty} f(n)$ converges iff $\int_1^{\infty} f(x)\,dx$ converges, when f is:

a. any function

b. bounded

c. increasing

d. **positive, decreasing, continuous** ✓

Applying Sequences and Series principles, Integral test conditions. Therefore, the verified correct option is positive, decreasing, continuous.

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44 [Challenging] • Target: Choosing the right test • Cross-Lesson: 12.02 Power Series and Taylor Series

Using the power series $1/(1-x) = \sum x^n$, find $\sum n x^{n-1} =$

a. $1/(1-x)$

b. $1/(1-x)^2$ ✓

c. $x/(1-x)^2$

d. $1/(1+x)^2$

Applying Convergence Tests and Power Series principles, Differentiate geometric series. Therefore, the verified correct option is $1/(1-x)^2$.

45 [Challenging] • Target: The limit comparison test • Cross-Lesson: 13.01 Vectors and Vector-Valued Functions

The series $\sum_{n=0}^{\infty} r^n$ converges for $|r| < 1$ to:

a. $1/(1-r)$ ✓

b. $r/(1-r)$

c. $1/(1+r)$

d. r

Applying Convergence Tests and Power Series principles, Geometric series sum $= 1/(1-r)$. Therefore, the verified correct option is $1/(1-r)$.

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